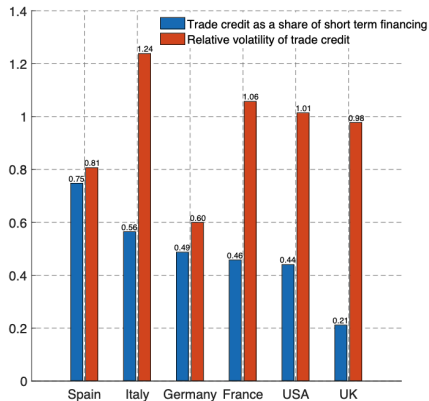


Macroeconomics of Trade Credit

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Presented by Mahdi Ebsim

Introduction



- ▶ **Bank Debt:** Borrowing from financial institutions
- ▶ **Trade Credit:** Firms can negotiate payment terms with suppliers

This paper:

- ▶ Introduces a model with trade and bank credit
- ▶ How do financial shocks propagate?
- ▶ Quantify the contribution to volatility due to trade credit with data

Preview

(1) Model with intra-period trade credit and bank debt

- ▶ Final good firms produce in the morning, collect sales in the afternoon
- ▶ Pay for intermediate goods in the morning, some production and/or sales in the afternoon
- ▶ Bank debt limited by borrowing constraint
 - Can only borrow up to a fraction $1 - \theta(s_t)$ of revenues in the afternoon
- ▶ Trade credit limited by reputation
 - Assume: If downstream defaults on payment, relationship severed permanently
 - Trade credit are afternoon revenues for suppliers $\Rightarrow$ relax their borrowing constraint

(2) Relative to a world without trade credit ...

- ▶ Trade credit expands credit supply $\Rightarrow$ increase output
- ▶ If suppliers are constrained, then trade credit amplifies credit shocks ($\searrow \theta(s_t)$)
- ▶ Calibrating model to data: Find that trade credit amplified output losses by $\approx 45\%$ in GFC

Order of Business

- ▶ Introduce Environment
- ▶ Simplified Economy
- ▶ Quantification

Model: Outline

- ▶ Discrete Time: $t = 0, 1, 2 \dots$
- ▶ Aggregate state: $s^t = (s_0, \dots, s_t)$ with $\pi(s_{t+1}|s_t)$
- ▶ Representative Household
 - Supply labour, purchase final good
- ▶ Continuum of **competitive** final good (*downstream*) firms
 - Face delay between production and receipt of revenues
- ▶ Finite number of **monopolistic** intermediate good (*upstream/supplier*) firms
- ▶ Competitive banks provide credit

Model: Household

$$U = \mathbb{E} \left\{ \sum_{t=0}^{\infty} \beta^t \left[C_t - \chi \frac{L_t^{1+\psi}}{1+\psi} \right] \right\}$$

$$\int p_{i,t} y_{i,t} di = W_t L_t + \Pi_t$$

$$C_t = \left(\int y_{i,t}^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}$$

$$y_{i,t} = \left(\frac{p_{i,t}}{P_t} \right)^{-\gamma} C_t$$

$$\chi L_t^{\psi} = W_t$$

Model: Firms

Final Good

- ▶ Endowed with capital k
- ▶ Buys intermediate goods $\{x_{i,j,t}\}_j$ from N_i intermediate good firms
- ▶ Technology

$$y_{i,t} = k^{1-\eta_i} \left\{ \left[\sum_{j=1}^{N_i} x_{i,j,t}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \right\}^{\eta_i}$$

- σ – elasticity of substitution across intermediate goods
- η_i – intermediate input share of production
- Assume $\sigma \geq \gamma$, intermediate inputs more substitutable than final goods

Intermediate Good

- ▶ Monopolists
- ▶ Linear technology $x_{i,j,t} = \ell_{i,j,t}$
- ▶ Call $(y_{i,t}, x_{i,j,t})$ a production line

Model: Financial Markets

Liquidity Problems

- ▶ Morning: Produce $\delta \in [0, 1]$ of $y_{i,t}$
- ▶ Afternoon: Receive $\pi_i \in [0, 1]$ of morning sales in the afternoon

Trade Credit

- ▶ Upstream makes take-it-or-leave-it offer $\{x_{i,j}(s^t), p_{i,j}^s(s^t), p_{i,j}^{tc}(s^t)\}$
 - Goods: $x_{i,j}(s^t)$
 - Morning payment: $p_{i,j}^s(s^t)$
 - Afternoon payment: $p_{i,j}^{tc}(s^t)$

Bank Debt

- ▶ Firms borrow $b_i(s^t)$ and $b_{i,j}(s^t)$
- ▶ Firms do not have ability to commit to repayment
 - In case of default; suffer a penalty equal to $(1 - \theta(s_t))$ of afternoon revenues

$$b_i(s^t) \leq (1 - \theta(s_t)) \underbrace{(1 - \delta(1 - \pi_i))}_{\text{afternoon. share}} \underbrace{p_i(s^t)y_i(s^t)}_{\equiv \text{rev}_i(s^t)}$$

$$b_{i,j}(s^t) \leq (1 - \theta(s_t))p_{i,j}^{tc}(s^t)$$

Model: Financial Markets

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$$\max \left\{ \sum_{j=1}^{N_i} p_{i,j}^s(s^t) - \delta(1 - \pi_i)rev_i(s^t), 0 \right\} \leq (1 - \theta(s_t))(1 - \delta(1 - \pi_i))rev_i(s^t)$$

$$\max \{ W(s^t)x_{i,j}(s^t) - p_{i,j}^s(s^t), 0 \} \leq (1 - \theta(s_t))p_{i,j}^{tc}(s^t)$$

Model: Trade Credit

- ▶ For a particular product line, fix $\{x_{i,j}(s^t), p_{i,j}^s(s^t), p_{i,j}^{tc}(s^t)\}$
- ▶ $J_i(s^t) \equiv$ discounted value of the final good firm under this contract

$$J_i(s^t) = \mathbb{E}_t \left\{ \sum_{\tau=t}^{\infty} \beta^{\tau-t} \left[rev_i(s^\tau) - \sum_{j=1}^{N_i} (p_{i,j}^s(s^\tau) + p_{i,j}^{tc}(s^\tau)) \right] \right\}$$

- ▶ $J_i^{(-j)}(s^t) \equiv$ discounted value of the final good firm under this contract except lose j forever

$$J_i^{(-j)}(s^t) = \mathbb{E}_t \left\{ \sum_{\tau=t}^{\infty} \beta^{\tau-t} \left[rev_i^{(-j)}(s^\tau) - \sum_{j' \neq j} (p_{i,j'}^s(s^\tau) + p_{i,j'}^{tc}(s^\tau)) \right] \right\}$$

- ▶ In symmetric equilibrium: $rev_i^{(-j)} = \left(\frac{N_i-1}{N_i} \right)^{\eta_i \frac{\sigma}{\sigma-1} \frac{\gamma-1}{\gamma}} rev_i$
- ▶ Define “surplus”: $\tilde{J}_i(s^t) = J_i(s^t) - J_i^{(-j)}(s^t)$
- ▶ No incentive to default if $p_{i,j}^{tc}(s^t) \leq \beta \mathbb{E}_t [\tilde{J}_i(s^{t+1})]$

Model: Suppliers Problem

$$\max_{p_{i,j}^s, p_{i,j}^{tc}, x_{i,j}} \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t [p_{i,j}^s(s^t) + p_{i,j}^{tc}(s^t) - W(s^t)x_{i,j}(s^t)] \right\}$$

subject to

$$b_i(s_t) \leq [1 - \theta(s_t)][1 - \delta(1 - \pi_i)]rev_i(s^t)$$

Downstream Borrowing

$$b_{i,j}(s_t) \leq [1 - \theta(s_t)]p_{i,j}^{tc}(s^t)$$

Upstream Borrowing

$$p_{i,j}^{tc}(s^t) \leq \beta \mathbb{E}[\tilde{J}_i(s^{t+1})|s^t]$$

Trade Credit IC

$$\tilde{J}_i(s^{t+1}) \geq 0 \quad \forall s^{t+1}$$

Trade Credit PC

$$0 \leq rev_i(s^t) - \sum_{j=1}^{N_i} (p_{i,j}^s(s^t) + p_{i,j}^{tc}(s^t)) \quad \forall s^t$$

Feasibility

- ▶ Internalize: $y_i(s^t)$ and $p_i(s^t)$
- ▶ Take as given aggregate quantities and trade credit of other suppliers

Model: Suppliers Problem

$$\max_{p_{i,j}^s, p_{i,j}^{tc}, x_{i,j}} \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t [p_{i,j}^s(s^t) + p_{i,j}^{tc}(s^t) - W(s^t)x_{i,j}(s^t)] \right\}$$

subject to

$$p_{i,j}^s(s^t) + \sum_{j' \neq j} p_{i,j'}^s(s^t) \leq [1 - \theta(s_t)][1 - \delta(1 - \pi_i)]rev_i(s^t)$$

Downstream Borrowing

$$W(s^t)x_{i,j}(s^t) - p_{i,j}^s(s^t) \leq [1 - \theta(s_t)]p_{i,j}^{tc}(s^t)$$

Upstream Borrowing

$$p_{i,j}^{tc}(s^t) \leq \beta \mathbb{E}[\tilde{J}_i(s^{t+1})|s^t]$$

Trade Credit IC

$$\tilde{J}_i(s^{t+1}) \geq 0 \quad \forall s^{t+1}$$

Trade Credit PC

$$0 \leq rev_i(s^t) - \sum_{j=1}^{N_i} (p_{i,j}^s(s^t) + p_{i,j}^{tc}(s^t)) \quad \forall s^t$$

Feasibility

Order of Business

- ▶ Introduce Environment
- ▶ Simplified Economy
- ▶ Quantification

Model: Simple Model

- ▶ $N_i = 1, k = 1$, common η
- ▶ Set $\delta = 0$ or $\pi_i = 1$
- ▶ Set $\psi = 0 \Rightarrow W = \chi$
- ▶ Equilibrium can be characterized by monopolist's problem
- ▶ Define *spot-economy* as one with $p^{tc} = 0$
 - Monopolist specifies $\{x(s^t), p^s(s^t)\}$
 - Solution:

$$x^{spot} = \left[\frac{\eta(1-\theta)}{W} \right]^{\frac{1}{1-\eta}}$$

- If $\theta = 0$, $w = \eta x^{\eta-1}$ and monopolist profit rate $(1 - \eta)$

Model: Simple Model

$$V(J, \theta) = \max_{x, p^s, p^{tc}, J'(\theta')} p^s + p^{tc} - Wx + \beta \mathbb{E}[V(J'(\theta'), \theta' | J, \theta)]$$

subject to

$$p^s \leq (1 - \theta)x^\eta \quad [\mu]$$

$$Wx - p^s \leq (1 - \theta)p^{tc} \quad [\iota]$$

$$p^{tc} \leq \beta \mathbb{E}[J'(\theta') | s^t] \quad [\kappa]$$

$$J'(\theta') \geq 0 \quad \forall \theta' \quad [\zeta]$$

$$0 \leq x^\eta - p^s - p^{tc} \quad \forall s^t \quad [\omega]$$

$$J \leq x^\eta - (p^s + p^{tc}) + \beta \mathbb{E}[J'(\theta') | J, \theta] \quad \forall s^t \quad [\lambda]$$

Promise Keeping

Model: Simple Model

$$x = \left[\frac{\eta(1 - \mu\theta + \iota)}{W(1 + \iota)} \right]^{\frac{1}{1-\eta}}$$

$$1 + \iota = \lambda + \omega + \mu$$

FOC of p^c

$$1 + \iota(1 - \theta) = \lambda + \kappa + \omega$$

FOC of p^{tc}

$$\lambda'(\theta') = \kappa + \lambda$$

FOC of $J'(\theta')$

- ▶ If supplier is unconstrained ($\iota = 0$)
 - Then $\mu = \kappa$, i.e. cost of banking and trade credit constraint equal
 - Supplier indifferent between spot and trade credit payments
 - As $\iota > 0$, trade credit becomes more costly
- ▶ $\lambda'(\theta') = \kappa + \lambda$ describes l.o.m for promise-keeping
 - If $\kappa > 0$, value of promise keeping rises
 - Binding trade-credit constraint pushes supplier to relax future credit constraints

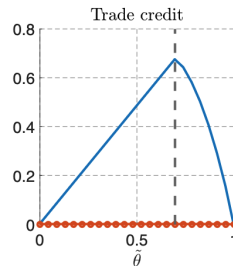
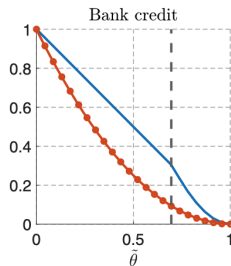
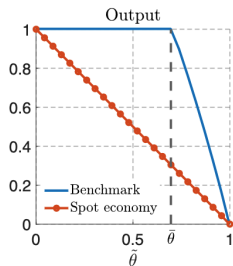
Model: Steady State

- Consider steady state: θ is fixed
- Let $\bar{\theta}$ be solution to $\eta = (1 - \bar{\theta})(1 + \beta\bar{\theta})$
- If $\theta \leq \bar{\theta}$, then

$$x = \left[\frac{\eta}{W} \right]^{\frac{1}{1-\eta}} \quad p^s = (1 - \theta)x^\eta \quad p^{tc} = \beta\theta x^\eta \quad J = \theta x^\eta$$

- If $\theta > \bar{\theta}$, then

$$x = \left[\frac{(1 - \theta)(1 + \beta\theta)}{W} \right]^{\frac{1}{1-\eta}} \quad p^s = (1 - \theta)x^\eta \quad p^{tc} = \beta\theta x^\eta \quad J = \theta x^\eta$$



Model: Response to Financial Shocks

► $\theta \in \{\tilde{\theta}, \tilde{\theta} + \varepsilon\}$ where $\varepsilon > 0$ and evolves according to $p(\theta'|\theta)$

► **Dampening:** If $\tilde{\theta} + \varepsilon \leq \bar{\theta}$, no output response

→ Trade credit increases to substitute for bank credit such that output remains unchanged

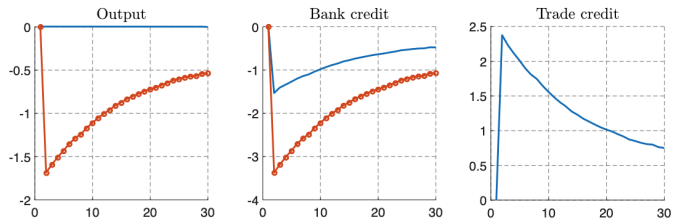
► **General:** If $\tilde{\theta} + \varepsilon > \bar{\theta}$, let $\varepsilon_{x,\theta} \equiv \frac{\partial \log x}{\partial \log \theta}$ and $\varepsilon_{p^{tc},\theta} \equiv \frac{\partial \log p^{tc}}{\partial \log \theta}$

$$\varepsilon_{x,\theta} = \alpha \varepsilon_{x,\theta}^{\text{spot}} + \frac{p^{tc}}{p^{tc} + (1 - \eta)x^\eta} \varepsilon_{p^{tc},\theta}$$
$$\varepsilon_{x,\theta}^{\text{spot}} = -\frac{\theta}{1 - \theta} \frac{1}{1 - \eta}$$

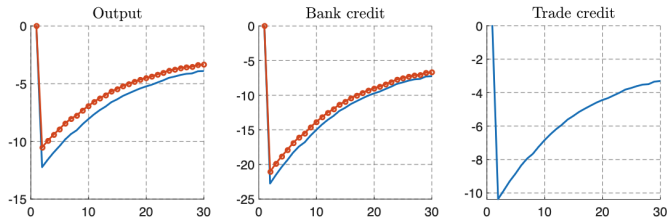
→ Amplification or dampening relative to spot-economy will be dependent on $\varepsilon_{p^{tc},\theta}$

→ If $\varepsilon_{p^{tc},\theta} < 0$, trade credit amplifies financial shocks

Model: Response to Financial Shocks



(a) Dampening of financial shocks



(b) Amplification of financial shocks

Full Model: Deterministic Steady State

- ▶ Similar to simple model, now with two cut-offs $\underline{\theta}_i$ and $\bar{\theta}_i$
- ▶ If $\theta < \underline{\theta}_i$

$$x_i = x_i^{unc} \quad p_i^{tc} = 0$$

$$p_i^s = N_i \left[1 - \frac{N_i - 1}{N - i} \eta_i^{\frac{\sigma}{\sigma-1} \frac{\gamma-1}{\gamma}} \right] rev_i(x_i^{unc})$$

- ▶ If $\theta \in (\underline{\theta}_i, \bar{\theta}_i]$

$$x_i = x_i^{unc} \quad p_i^s = (1 - \theta(1 - \delta + \delta\pi_i)) rev_i(x_i^{unc})$$

$$p_i^{tc} = \beta N_i \left\{ \left[1 - \frac{N_i - 1}{N_i} \eta_i^{\frac{\sigma}{\sigma-1} \frac{\gamma-1}{\gamma}} \right] - \frac{1}{N_i} [1 - \theta(1 - \delta + \delta\pi_i)] \right\} rev_i(x_i^{unc})$$

- ▶ If $\theta > \bar{\theta}_i$

$$x_i = x_i^{con} \quad p_i^s = (1 - \theta(1 - \delta + \delta\pi_i)) rev_i(x_i^{con})$$

$$p_i^{tc} = \beta N_i \left\{ \left[1 - \frac{N_i - 1}{N_i} \eta_i^{\frac{\sigma}{\sigma-1} \frac{\gamma-1}{\gamma}} \right] - \frac{1}{N_i} [1 - \theta(1 - \delta + \delta\pi_i)] \right\} rev_i(x_i^{con})$$

Predictions

- ▶ Model makes some predictions for the cross-section
- ▶ If firms have high intermediate input share $\Rightarrow$ trade credit as a share of sales increases

$$\frac{\partial \frac{p_i^{tc}}{rev_i}}{\partial \eta_i} > 0$$

- ▶ If firms have large accounts receivables balance $\Rightarrow$ trade credit as a share of sales increases

$$\frac{\partial \frac{p_i^{tc}}{rev_i}}{\partial \pi_i} > 0$$

- ▶ If firms supply from concentrated markets $\Rightarrow$ trade credit as a share of sales increases

$$\frac{\partial \frac{p_i^{tc}}{rev_i}}{\partial N_i} < 0$$

$\rightarrow \sigma \geq \gamma$ sufficient to generate sign

$\rightarrow$ If there are very few firms: $\tilde{J}$ increases $\rightarrow p^{tc}$ limit increases

Order of Business

- ▶ Introduce Environment
- ▶ Simplified Economy
- ▶ Quantification

- ▶ ORBIS, annual balanced panel of non-financial firms between 2007-2015 in Italy
- ▶ Observe: revenues, sales, short-term loans, intermediate good expenditures, accounts payable and receivables.
- ▶ Aggregate into 58 sectors and map the following to the model:

- $\frac{\text{Accounts Payable}}{\text{Sales}} = \frac{p_{i,t}^{tc}}{rev_{i,t}}$
- $\frac{\text{Intermediate Expenditures}}{\text{Sales}} = \frac{p_{i,t}^s + p_{i,t}^{tc}}{rev_{i,t}}$
- $\frac{\text{Accounts Receivables}}{\text{Sales}} = \delta \pi_i$
- $\frac{\text{Short-term Bank Debt}}{\text{Sales}} = 1 - \theta_t(1 - \delta + \delta(1 - \pi_i))$
- $HHI_{supplier} = \frac{1}{N_i}$

Testing Cross-sectional Predictions

- Model predicts $\frac{\partial \frac{p_i^{tc}}{rev_i}}{\partial \eta_i} > 0$, $\frac{\partial \frac{p_i^{tc}}{rev_i}}{\partial \pi_i} > 0$, and $\frac{\partial \frac{p_i^{tc}}{rev_i}}{\partial N_i} < 0$

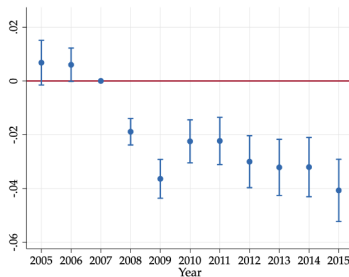
Table 2: Cross-sectional sector-level regressions

	<i>Dep. variable: Accounts payable/sales</i>			
	(1)	(2)	(3)	(4)
Accounts receivable/sales	0.297*** (0.029)			0.326*** (0.027)
Intermediate inputs /sales		0.154*** (0.028)		0.239*** (0.020)
HHI ^{supplier}			0.655*** (0.111)	0.303*** (0.085)
Adj. R^2	0.329	0.092	0.116	0.477
Obs.	522	522	522	522

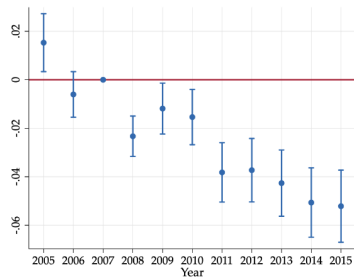
Testing Cross-sectional Predictions

- ▶ Model predicts final good firms with comparatively more constrained suppliers reduce actively more
- ▶ Compute for each sector $\bar{\theta}_i$, threshold value for constraints in 2007
- ▶ Estimate on firm panel the following

$$y_{f,i,t} = \alpha_f + \beta_t \times \mathbb{1} [\bar{\theta}_i < \text{median}(\bar{\theta}_i)]$$



(a) Sales

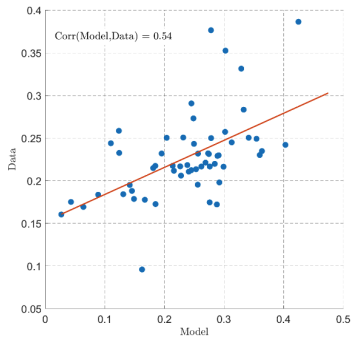


(b) Accounts payable

- Fixed Parameters: $\beta = 0.98$, $\psi = 1.00$, and $p(\theta_L|\theta_L) = 0.99$

Table 3: Model parameters and targeted moments

Parameter	Value	Moment	Data	Model
Mean(π_i)	0.52	Distribution of accounts receivable/sales		
Stdev(π_i)	0.14			
Mean(η_i)	0.81	Distribution of intermediate inputs/sales		
Stdev(η_i)	0.09			
Mean(N_i)	23.83	Distribution of HHI ^{supplier}		
Stdev(N_i)	10.96			
δ	0.55	Mean(Accounts payable/sales)	0.23	0.23
χ	3.50	Mean(Worked hours/total hours)	0.33	0.33
θ_L	0.78	Mean(Bank loans/sales) in 2007	0.16	0.16
θ_H	0.84	Mean(Bank loans/sales) in 2008	0.12	0.12
$p(\theta_H \theta_H)$	0.86	Mean(Bank loans/sales) in 2011	0.14	0.14
γ	3.00	% Fall in average firms' sales	0.12	0.12
σ	5.00	β_{2008} in eq. (22), log-sales as dep. var.	-0.02	-0.02



(a) Trade credit across sectors

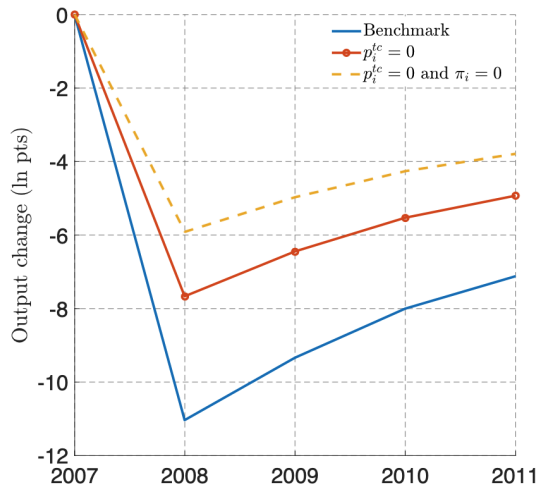
	Data	Model
$\text{corr}(\text{payable}/\text{sales}, \text{receivable}/\text{sales})$	0.59	0.48
$\text{corr}(\text{payable}/\text{sales}, \text{intermediate-inputs}/\text{sales})$	0.21	0.68
$\text{corr}(\text{payable}/\text{sales}, \text{HHI}^{\text{supplier}})$	0.24	0.30
$\text{corr}(\text{payable}/\text{sales}, \text{bank debt}/\text{sales})$	0.35	0.59

(b) Cross-sector correlation patterns

Credit Multiplier

	Benchmark	$p_i^{\text{tc}} = 0$	$p_i^{\text{tc}} = 0$ and $\pi_i = 0$
Bank credit	1.00	0.43	0.33
To final good firms	0.73	0.43	0.33
To suppliers	0.27	0.00	
% allocated to wages	0.96	0.02	0.03
Output	1.00	0.60	0.86

Crisis Episode



Conclusion

- ▶ Introduce a model with trade and bank credit
- ▶ Illustrate that trade credit provides significant liquidity
- ▶ Show trade credit is relevant for the propagation of aggregate financial shocks
- ▶ Depending on financial health of suppliers, trade credit can amplify or dampen financial shocks
- ▶ Some comments
 - Can you think about the financial consequences of the supply chain story?
 - $\sigma \geq \gamma$ seems arbitrary, shocks to suppliers will have more bite if $\sigma \rightarrow 1$